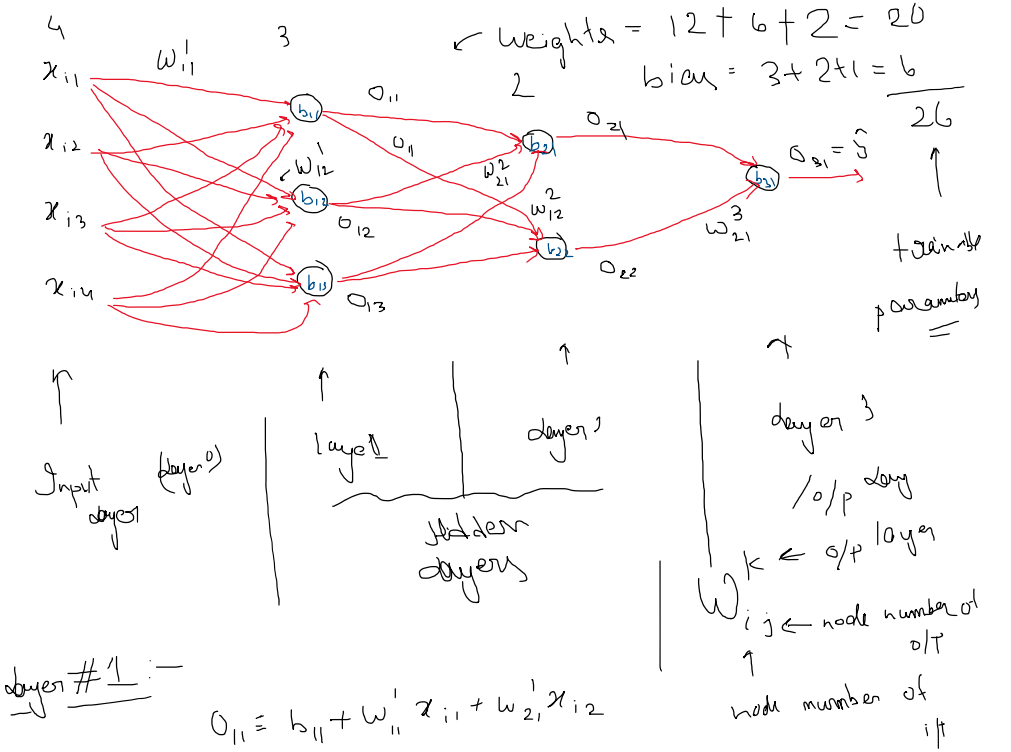


cgpa	iq	10 th Marks	12 th Marks	place d
7.2	72	492	315	0
9.64	148	608	436	1

Forward propagation is the process in a neural network where input data is passed through the network's layers to generate an output or prediction



layer #1 :-

$$o_{11} = b_{h1} + w_{11}x_{i1} + w_{21}x_{i2} + w_{31}x_{i3} + w_{41}x_{i4}$$

$$\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \\ w_{41} & w_{42} & w_{43} \end{bmatrix} \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix}$$

$(4 \times 3) \quad (1 \times 4)$

$$\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \\ w_{41} & w_{42} & w_{43} \end{bmatrix}^T \times \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{h1} \\ b_{h2} \\ b_{h3} \end{bmatrix} = \begin{bmatrix} o_{11} \\ o_{12} \\ o_{13} \end{bmatrix}$$

$(3, 4) \quad (4, 1) \quad (3, 1) \quad (3, 1)$

activation function $\hat{y} = ?$

$\hat{y} = \sigma(z)$ $\hat{y} = \tanh(z)$

layer 2

$$\begin{bmatrix} w_{11}^2 & w_{12}^2 \\ w_{21}^2 & w_{22}^2 \end{bmatrix} \begin{bmatrix} o_{11} \\ o_{12} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

- $\downarrow$ -

$$\begin{bmatrix} w_{11}^2 & w_{12}^2 \\ w_{21}^2 & w_{22}^2 \\ w_{31}^2 & w_{32}^2 \end{bmatrix} \begin{bmatrix} 0_{11} \\ 0_{12} \\ 0_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

(2,3) (3,1) (2,1)

$$\begin{bmatrix} z_{11} \\ z_{12} \end{bmatrix} \xrightarrow{\sigma} \begin{bmatrix} o_{21} \\ o_{22} \end{bmatrix}$$

Layer 3

$$\begin{bmatrix} w_{11}^3 \\ w_{21}^3 \end{bmatrix} \begin{bmatrix} o_{21} \\ o_{22} \end{bmatrix} + [b_{31}] = [z_{31}] \xrightarrow{\sigma} [o_{31}]$$

$$a^{[1]} = \sigma \left(a^{[0]} w^{[1]} + b^{[1]} \right)$$

Prod $\downarrow$

$$\sigma \left(\sigma \left(\frac{\sigma(a^{[0]} w^{[1]} + b^{[1]})}{a^{[1]}} \times w^{[2]} + b^{[2]} \right) \times w^{[3]} + b^{[3]} \right)$$

$a^{[2]}$

$a^{[3]}$